



Affirmed Networks UnityCloud Deployment Guide for KVM

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Affirmed Networks Inc.

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Contents

Preface	vi
Audience	vi
Release Information	vi
How to Obtain Product Documentation.....	vi
Customer Support.....	vi
Introduction	1
Prerequisites	1
Script Installation	2
Install and Unpack Scripts	2
OPM Creation and Configuration	4
Create the OPM	4
Next Steps.....	8
UC Cluster Creation and Configuration.....	9
Create the UC Cluster.....	9
Next Steps.....	15

List of Figures

	Example System Layout	1
	Check System Information – Example CLI Output	3
	Generating the JSON file: sample output	4
Figure 1.	Using RH Cockpit to Monitor VM Boot	7
Figure 2.		
Figure 3.	Using RH Cockpit to Monitor VM boot	15
Figure 4.		
Figure 5.		

List of Tables

Table 1.	<vm_name>.json File Parameters for OPM Cluster Configuration (VM Section).....	5
Table 2.	<vm_name>.json File Parameters for OPM Cluster Configuration (XML Section).....	5
Table 3.	Multimaster and SR-IOV Worker Example Setup	9
Table 4.	<vm_name>.json File Parameters for UC Cluster Configuration (VM Section).....	10
Table 5.	<vm_name>.json File Parameters for UC Cluster Configuration (XML Section).....	11

Preface

This document describes how to install UnityCloud on the KVM server.

Audience

This document is intended for network operators (sometimes referred to as Users or Operators in this document), installation staff, field engineers, support engineers, and other qualified service personnel who want to install UnityCloud in a RHEL9.2-KVM server environment.

Release Information

See the Release Notes for important information, requirements, limitations, and restrictions that apply to this release.

How to Obtain Product Documentation

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Introduction

This guide describes how to install UnityCloud software on a KVM server.

Prerequisites

Complete the following tasks before installing UnityCloud software.

1. Download the UnityCloud software image. Contact your Support team to obtain the latest UC software image.
2. Download the `uc_conf.py` script and templates (`vm-kvm_uc_onboard-0.x.x-*.tgz` package). Figure 1 shows an example system layout.

Note: Contact the Support team to obtain the scripts if they are not available on the software repository. Ask for `vm-kvm_uc_onboard-0.2.14-*.tgz` (or higher version).

Example System Layout

Figure 1.

```
[root@srv1 uc] # pwd
/kvm/uc
[root@srv1 uc] # ls -lR
.:
total 66
-rw-r--r-- 1 root root 3597 Jun 6 10:40 pci_info.sh
drwxr-xr-x 2 root root 384 Jun 6 10:40 templates
-rwxr-xr-x 1 root root 62326 Jun 8 17:02 uc_conf.py

./templates:
total 34
-rw-r--r-- 1 root root 5173 Jun 6 10:40 multimaster.template.xml
-rw-r--r-- 1 root root 4720 Jun 6 10:40 opm.template.xml
-rw-r--r-- 1 root root 4965 Jun 6 10:40 singlemaster.template.xml
-rw-r--r-- 1 root root 5832 Jun 6 10:40 worker4sriov.template.xml
-rw-r--r-- 1 root root 6685 Jun 6 10:40 worker8sriov.template.xml
-rw-r--r-- 1 root root 4972 Jun 6 10:40 workernonsriov.template.xml
```

Script Installation

This section describes how to install scripts required to deploy UnityCloud in a KVM environment.

Install and Unpack Scripts

Use the following procedure to unpack and install required deployment scripts.

1. Log in to the server as the root user.
2. Create the `/kvm/uc` directory on the RHEL9 hypervisor that you created using the procedures described in *Affirmed Networks RHEL 9.2 Installation and Configuration for KVM: System Preparation Guide*.
3. Copy the `vm-kvm_uc_onboard-0.x.x-*.tgz` package to the `/kvm/uc` directory and unpack it there (as shown in Figure 1):

```
tar -xf <tgz file>
```

4. Upload the `vm-opm*.qcow2` and `vm-platform*.qcow2` images to `/kvm/uc`.
5. Set the script permissions:

```
chmod 770 uc_conf.py
```

```
chmod 770 pci_info.sh
```

6. Check the system information. This helps planning VMs layout, and creating the json configuration files for VMs:

```
./uc_conf.py --sysinfo
```

Figure 2 shows sample output with UC VMs already created on the hypervisor. Available vCPUs that are dedicated to VMs are shown in **green**, and vCPUs already assigned to running VMs are shown in **red**. Entries that appear in **red** with an asterisk(*) indicate vCPUs assigned to UC VMs loaded to the config store but not yet created.

Check System Information – Example CLI Output

Figure

```

SYSTEM INFO:
Total Memory: 754GB
Total 1G Hugepages: 710
NUMA0 Free 1G hugepages: 269
NUMA1 Free 1G hugepages: 191

NUMA nodes: 2
NUMA node0: 0 2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50 52 54 56 58 60 62 64 66 68 70 72 74 76 78
NUMA node1: 1 3 5 7 9 11 13 15 17 19 21 23 25 27 29 31 33 35 37 39 41 43 45 47 49 51 53 55 57 59 61 63 65 67 69 71 73 75 77 79

Kernel Isolated CPUs: 4-39,44-79
CPU Partitioning Dedicated CPUs: 4-39,44-79

CPU siblings list:

NUMA0:
-----
CPU:      0  2  4  6  8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38
Sibling: 40 42 44 46 48 50 52 54 56 58 60 62 64 66 68 70 72 74 76 78

CPU:      40 42 44 46 48 50 52 54 56 58 60 62 64 66 68 70 72 74 76 78
Sibling:  0  2  4  6  8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38

NUMA1:
-----
CPU:      1  3  5  7  9 11 13 15 17 19 21 23 25 27 29 31 33 35 37 39
Sibling: 41 43 45 47 49 51 53 55 57 59 61 63 65 67 69 71 73 75 77 79

CPU:      41 43 45 47 49 51 53 55 57 59 61 63 65 67 69 71 73 75 77 79
Sibling:  1  3  5  7  9 11 13 15 17 19 21 23 25 27 29 31 33 35 37 39

Interfaces per NUMA node:
NUMA0: eno1 (igb), eno2 (igb), eno3 (igb), eno4 (igb), ens1f0 (ice), ens1f1 (ice), ens3f0np0 (mlx5_core), ens3f1np1 (mlx5_core)
NUMA1: ens4f0 (ice), ens4f1 (ice), ens5f0 (i40e), ens5f1 (i40e), ens7f0 (i40e), ens7f1 (i40e), ens8f0np0 (mlx5_core), ens8f1np1 (mlx5_core)

```

OPM Creation and Configuration

This section describes how to create the OPM and configure required parameters.

Create the OPM

Use the following procedure to create and configure the OPM.

1. Generate the JSON file using the opm template:

```
# ./uc_conf.py --generate /kvm/uc/templates/opm.template.xml
```

Generating the JSON file: sample output

Figure 3.

```
-----
Running as ROOT user - PASSED
-----
Supported Version - PASSED
-----

Generated from templates/opm.template.xml:

{
  "vm": {
    "name": "",
    "vm_type": "opm",
    "vcpus": "",
    "vcpus_list": "",
    "memory": "",
    "payload": "",
    "size": "",
    "seconddisksize": "",
    "ctrl_network": ""
  },
  "xml": {
    "ff_k8sversion": "none",
    "ctrlp_ip": "",
    "ctrlp_netmask": "",
    "ctrlp_gw": "",
    "hostname": "",
    "dns1": "",
    "dns2": "",
    "domain": "",
    "ntp": "",
    "timezone": "UTC",
    "vnfmgr_dreg_external_port": "5005",
    "k8_master_token": ""
  }
}
```

2. Create the <vm_name>.json file and use the generated JSON information as the content base.

OPM Creation and Configuration

3. Fill in all required fields with data, as shown in the following tables.

Table 1. <vm_name>.json File Parameters for OPM Cluster Configuration (VM Section)

Parameter	Description	Mandatory or Optional
name	Name of the VM as used by virsh	Mandatory
vm_type	One of the VM types: opm, k8-master, k8-node	Mandatory
vcpus	Number of vCPUs assigned to the VM	Mandatory
vcpus_list	List of vCPUs that will be pinned to the VM, siblings from the same Numa should be provided. A value of 'numa0' will automatically assign vCPUs from that numa on loading configuration.	Mandatory
memory	Integer value representing MB.	Mandatory
payload	Location of the qcow2. Enter the full path.	Mandatory
size	Size of the VM disk provided in GiB. Example: 150G. For UC 5.x if ff_in_service_upgrade is set to True this size needs to be double the size of what you want the disk size to be.	Mandatory
seconddisksize	Provide as xxxG, second disk for /filestore. If configured, it should be configured for all VMs in the OPM cluster. It is preferred to have a second disk. Remove the line, if not needed.	Optional - remove the line if not used
ctrl_network	The bridge network. Format as bridge:bridge0 , where bridge0 is the name of the bridge that the VM will use. (If VLAN tagging is needed, use the tagged bridge created on the system).	Mandatory

Table 2. <vm_name>.json File Parameters for OPM Cluster Configuration (XML Section)

Parameter	Description	Comments
ff_k8sversion	Feature flag of the k8s version to be used. The default "none" means that there is no override and the default version is used.	
ff_in_service_upgrade	Feature flag to enable VM ISSU feature. This enables VM ISSU feature which creates an addition disk partition for ISSU. This means that the VM disk size needs to be double the size of what you want the disk size to be.	Default: "True" Applies to UC 5.x VMs, 4.x VMs will ignore this parameter.

OPM Creation and Configuration

Parameter	Description	Comments
ctrlp_ip	IP that will be configured on the VM using ctrl_network.	
ctrlp_netmask	Netmask of ctrl_ip	
ctrlp_gw	GW for the ctrlp_ip	
hostname	Hostname	
dns1	Primary DNS	
dns2	Secondary DNS	
domain	Domain	
ntp	NTP server IP	
timezone	Time zone for the guest OS	Default is UTC
vnfmgr_dreg_external_port	cnfmgr docker reg external port	Default is 5005
k8_master_token	Cluster token, same on all VMs belonging to the same cluster.	Generate: echo \$(head -c 20 /dev/urandom base64 tr -dc 'a-z0-9' head -c 6)". "\$(head -c 40 /dev/urandom base64 tr -dc 'a-z0-9' head -c 16)
mtu	Internal mtu	Default is 8950
enable_root		Default '1'
disable_maint	Maint group (non-root bash user(s)) are enabled by default	Default '0'
root_passwd	Replace default root sha512 password. <i>mkpasswd -m sha-512</i> can be used to generate the hash.	Default 'none' will set default password
root_pubkey	Define public key for root user authentication	Default none
zone	Zone configuration, same server will be always the same zone	Default 'kvm'

Notes:

- Make sure to enter all data correctly in the XML section. XML data is not validated.
- After the VM comes up, check the *configparser.log* to verify that configuration was applied successfully.

4. Using the JSON file and the template used to generate it, load the configuration into config store:

OPM Creation and Configuration

```
./uc_conf.py --load <vm_name>.json:templates/opm.template.xml
```

5. Enter the following command to verify the data in the store:

```
./uc_conf.py --dump
```

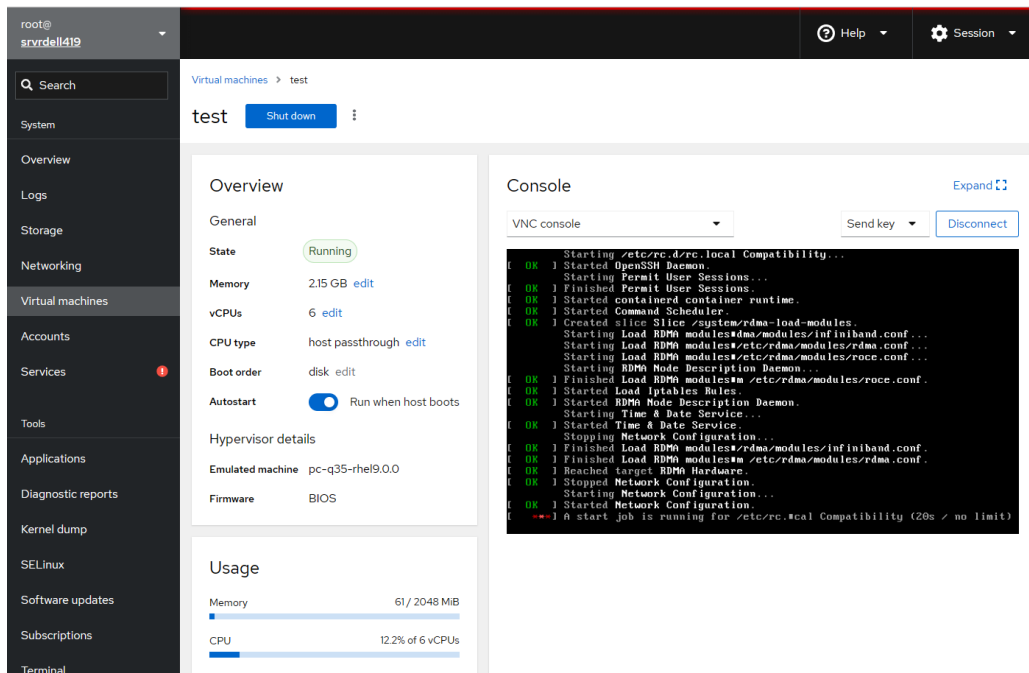
6. Create the VM:

```
./uc_conf.py --create <vm_name>
```

7. Using the RH Cockpit monitor the VM boot via VNC for errors:

Using RH Cockpit to Monitor VM Boot

Figure 4.



8. Inside the VM, check the `/var/log/configParser.log` file to verify that data provided in the XML section of the JSON file was handled correctly. The last line in the file should say “configParser - Completed.”

9. After the VM has finished booting up, log in and collect data required for the UC cluster (master and worker VMs):

- a. OPN dreg certificate. For a single OPM, enter:

```
sed ' :a;N;$!ba;s/\n/\\n/g' /vnfmgr/certs/dreg.crt
```

- b. For a Mult-OPM, enter:

```
kubectl -n vnfmgr exec -it vnfmgr-dreg-685f6c567f-2dfrc -- sed  
' :a;N;$!ba;s/\n/\\n/g' /certs/dreg.crt
```

- c. OPM m2m_pubkey. Enter:

```
cat /etc/ssh/ssh_host_ed25519_key.pub
```

Next Steps

Complete the procedures provided in the following sections of the *UnityCloud Deployment Guide* to prepare for cluster creation. Use these procedures as your starting points.

- Downloading the mongoDB Used in OPM.
- Attaching to an Existing Kubernetes Cluster
- Downloading Affirmed Networks Images

UC Cluster Creation and Configuration

This section describes how to create the UC cluster and configure required parameters.

Create the UC Cluster

1. Generate the JSON file using the master or worker template, depending on type of VM you are using.

Table 3. Multimaster and SR-IOV Worker Example Setup

Master	Worker
<pre>./uc_conf.py --generate templates/multimaster.template.xml</pre> <pre>----- Running as ROOT user - PASSED ----- Supported Version - PASSED ----- Generated from templates/multimaster.template.xml: { "vm": { "name": "", "vm_type": "k8-master", "vcpus": "", "vcpus_list": "", "memory": "", "payload": "", "size": "", "seconddisksize": "", "ctrl_network": "" }, "xml": { "ff_k8sversion": "none", "topo_mgr_pol": "none", "k8_master_ip": "", "k8_master_netmask": "", "k8_master_gw": "", "k8_master_primary": "", "k8extram_ip_1": "", "k8extram_ip_2": "", "k8_primary_master_ip": "", "k8_master_lb_vip": "", "k8_master_token": "", "k8_cert_key": "", "hostname": "", "dns": "", "domain": "", "timezone": "UTC", "ntp": "", "vnfmgr_dreg_external_ip": "", "vnfmgr_dreg_external_port": "5005", "vnfmgr_dreg_cert": "", "vnfmgr_m2m_pubkey": "" } }</pre>	<pre>./uc_conf.py --generate templates/worker4sriov.template.xml</pre> <pre>----- Running as ROOT user - PASSED ----- Supported Version - PASSED ----- Generated from templates/worker4sriov.template.xml: { "vm": { "name": "", "vm_type": "k8-node", "vcpus": "", "vcpus_list": "", "memory": "", "payload": "", "size": "", "seconddisksize": "", "rookdisksize": "", "ctrl_network": "", "sriov_1": "", "sriov_2": "", "sriov_3": "", "sriov_4": "" }, "xml": { "ff_k8sversion": "none", "reserve_cpu": "", "hugepages_enable": "false", "k8_master_ip": "", "k8_master_vip": "", "k8_master_token": "", "k8_worker_ip": "", "k8_worker_netmask": "", "k8_worker_gw": "", "mtu": "8950", "hostname": "none", "dns": "", "domain": "", "timezone": "UTC", "ntp": "", "vnfmgr_dreg_external_ip": "", "vnfmgr_dreg_external_port": "5005", "vnfmgr_dreg_cert": "", "vnfmgr_m2m_pubkey": "", "enable_root": "1", "disable_maint": "0", "root_passwd": "none", "root_pubkey": "none", "zone": "kvm", "sriov_1_net_id_label": "sr1", "sriov_2_net_id_label": "sr1", "sriov_3_net_id_label": "sr2", "sriov_4_net_id_label": "sr2", "sriov_1_net_id_dpdk": "True", "sriov_2_net_id_dpdk": "True", "sriov_3_net_id_dpdk": "True", "sriov_4_net_id_dpdk": "True" } }</pre>

UC Cluster Creation and Configuration

2. Create the `<vm_name>.json` file and use generated JSON information as the content base.
3. Complete required fields with the appropriate data, as shown in the following tables.

Table 4. `<vm_name>.json` File Parameters for UC Cluster Configuration (VM Section)

Parameter	Description	Mandatory or Optional
name	Name of the VM, used by virsh	Mandatory
vm_type	One of VM types: opm, k8-master, k8-node	Mandatory
vcpus	Number of vCPUs assigned to the VM	Mandatory
vcpus_list	List of vCPUs that will be pinned to the VM. Siblings from the same Numa should be provided. A value of 'numa0' will automatically assign vCPUs from that numa on loading configuration.	Mandatory
memory	Integer value representing MB	Mandatory
payload	Location of the qcow2. Enter the full path.	Mandatory
size	Size of the VM disk provided in GiB as xxxG For UC 5.x if ff_in_service_upgrade is set to True this size needs to be double the size of what you want the disk size to be.	Mandatory
seconddisksize	Provide as xxxG, second disk for /data, second disk to store pcaps and fluentd log. If configured, should be configured for all VMs. It's preferred to have a second disk. If not needed, remove the line. Mandatory if rookdisk is configured.	Optional - remove the line if not used
rookdisksize	Size of the disk for rook-ceph. In the case of a multi-opm, this is used for the storage disks. If ff_mm_opm_disk_size is non-zero, set format disk and mkdir /var/storage along with some sub dirs. If zero, then additional disk is just like before, used for rook-ceph.	Optional - remove the line if not used
ctrl_network	The bridge network: format as bridge:bridge0, where bridge0 is the name of the bridge that the VM will use. If VLAN tagging is needed, use the tagged bridge created on the system.	Mandatory

UC Cluster Creation and Configuration

Parameter	Description	Mandatory or Optional
sriov_1 to sriov_X,	sriov vf's/devices that will be attached to the VM. The number should match the template. Use pci_info.sh script to get the virsh naming format example: "sriov_1": "pci_0000_3b_01_0", All the sriov_X VFs that have the same label (xml section) should be from same sriov PF. Instead of providing pci_*, the PF interface can be provided to automatically assign the virtual function from that PF on loading the configuration.	Mandatory if sriov template used

Table 5. <vm_name>.json File Parameters for UC Cluster Configuration (XML Section)

Parameter	Description	Comments
ff_k8sversion	Feature flag of the k8s version to be used, default "none" means there is no override and default version is used.	
ff_in_service_upgrade	Feature flag to enable VM ISSU feature. This enables VM ISSU feature which creates an addition disk partition for ISSU. This means that the VM disk size needs to be double the size of what you want the disk size to be.	Default: "True" Applies to UC 5.x VMs, 4.x VMs will ignore this parameter.
ff_mm_opm_disk_size	For multimode OPM, this value for the worker node configures opm third disk for storing repository instead of rookdisk size. If ff_mm_opm_disk_size is non-zero, set format disk and mkdir /var/storage along with some sub dirs. If zero, then additional disk is just like before, used for rook-ceph.	Default "0"
reserve_cpu	Should be empty if not configured. On UPF set to "0,1,2"	Default is "" -disable reserve cpu but keeps affine of kube/system daemons to all cpus

UC Cluster Creation and Configuration

Parameter	Description	Comments
		UC 5.x adds option to set it to "9999" "9999" – disable reserve cpu and disable cpu manager and don't affine processes
hugepages_enabled	false by default, true for UPF	Default is "false"
topo_mgr_pol	"none" by default, in case of UPF cluster "single-numa-node"	Default is "none"
k8_master_ip	For Master nodes: IP assigned to this master on ctrl_network For worker nodes: IP of the primary master node	
k8_master_vip	Virtual IP of the masters cluster. For workers in the single master cluster this should be set to "none".	
k8_master_netmask	Netmask for this master IP on ctrl_network	
k8_master_gw	GW for this master IP on ctrl_network	
k8_worker_ip	IP assigned to this worker on ctrl_network	
k8_worker_netmask	Netmask for this worker IP on ctrl_network	
k8_worker_gw	GW for this worker IP on ctrl_network	
k8_master_primary	Set to 1 on primary master node, 0 on rest of the masters.	
k8_extram_ip_1	IP of second master. Same value on all master nodes.	

UC Cluster Creation and Configuration

Parameter	Description	Comments
k8_extram_ip_2	IP of third master. Same value on all master nodes.	
k8_primary_master_ip	Set to IP assigned to Primary master. Same value on all master nodes.	
k8_master_lb_vip	Virtual IP of the master's cluster.	
k8_master_token	Cluster token, same on all VMs belonging to the same cluster.	Generate: <pre>echo \$(head -c 20 /dev/urandom base64 tr -dc 'a-z0-9' head -c 6)"."\$(head -c 40 /dev/urandom base64 tr -dc 'a-z0-9' head -c 16)</pre>
k8_cert_key		Generate: <pre>openssl rand -hex 32</pre>
hostname	Hostname	
dns1	DNS	
domain	Domain	
ntp	NTP server IP	
timezone	Time zone for the guest OS.	Default is UTC
vnfmgr_dreg_external_ip	For a single OPM this is the mgmt IP address. For a multi OPM, use the istio external IP of vnfmgr (kc get svc -n vnfmgr).	
vnfmgr_dreg_external_port	cnfmgr docker reg external port	Default is 5005

UC Cluster Creation and Configuration

Parameter	Description	Comments
vnfmgr_dreg_cert	Use data from OPM Creation Step 9a.	
vnfmgr_m2m_pubkey	Use data from OPM Creation Step 9b.	
mtu	Internal mtu	Default is 8950
enable_root		Default '1'
disable_maint	Maint group (non-root bash user(s)) are enabled by default	Default '0'
root_passwd	Replace default root sha512 password <i>mkpasswd -m sha-512</i> can be used to generate the hash	Default 'none' will set default password
zone	Zone configuration, same server will be always the same zone.	Default 'kvm'
root_pubkey	Define public key for root user authentication	Default none
sriov_X_net_id_label	Label for the sriov device	Default labels provided when generating json from template
sriov_X_net_id_dpdk	Will the interface be dpdk	Default "True"

Notes:

- Make sure to enter all data correctly in the XML section. XML data is not validated.
- After the VM comes up, check the *configparser.log* to verify that configuration was applied successfully.

4. Using the JSON file and template used to generate it , load the configuration into config store:

```
./uc_conf.py --load <vm_name>.json:templates/<template used for this vm>
```

5. Enter the following command to verify the data in the store:

```
./uc_conf.py --dump
```

6. Repeat Steps 4 and 5 for each VM in the cluster.

UC Cluster Creation and Configuration

7. Create the VMs:

```
./uc_conf.py --create <vm_name>
```

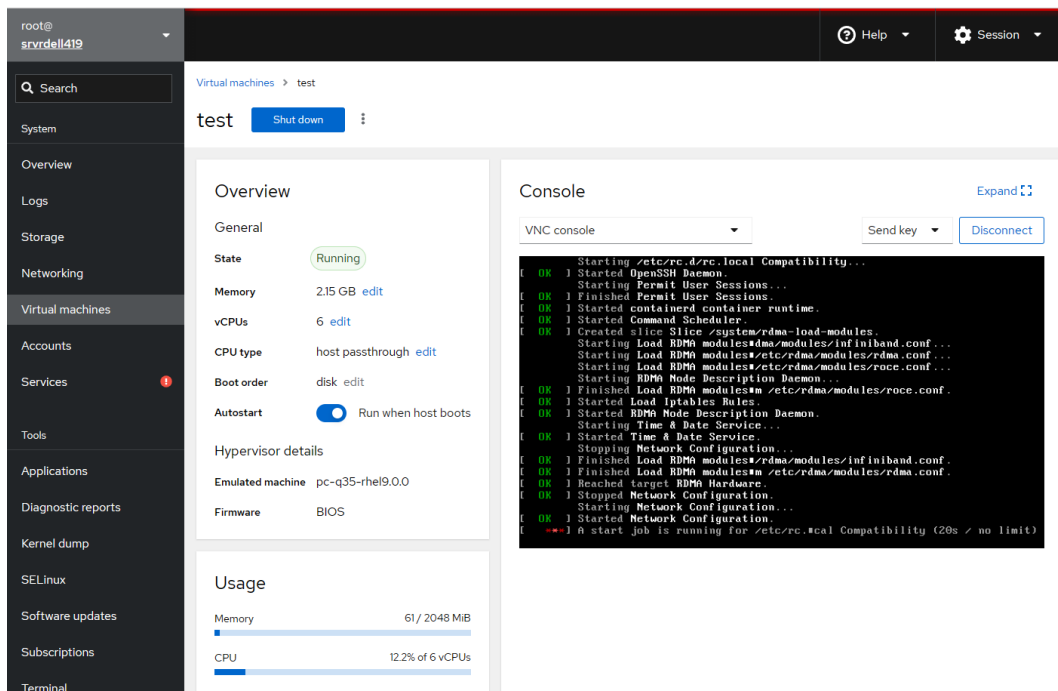
Note:

Create the VMs on all servers for the same cluster so that primary master is first. After one minute, create all other masters for the same UC cluster. Note that there is a two hour expiration timeout for **ctrplane**, masters creation. Workers can be added at any time afterwards.

8. Using the RH Cockpit monitor the VM boot via VNC for errors:

Using RH Cockpit to Monitor VM boot

Figure 5.



9. Inside the VM, check the `/var/log/configParser.log` file to verify that data provided in the XML section of the JSON file was handled correctly. The last line in the file should say “configParser - Completed.”

Next Steps

Complete the procedures provided in the following sections of the *UnityCloud Deployment Guide* to complete the deployment. Use these procedures as your starting points.

- Downloading the mongoDB Used in OPM
- Installing MetalLB Federation

- Deploying UnityCloud Network Function Applications With OPM